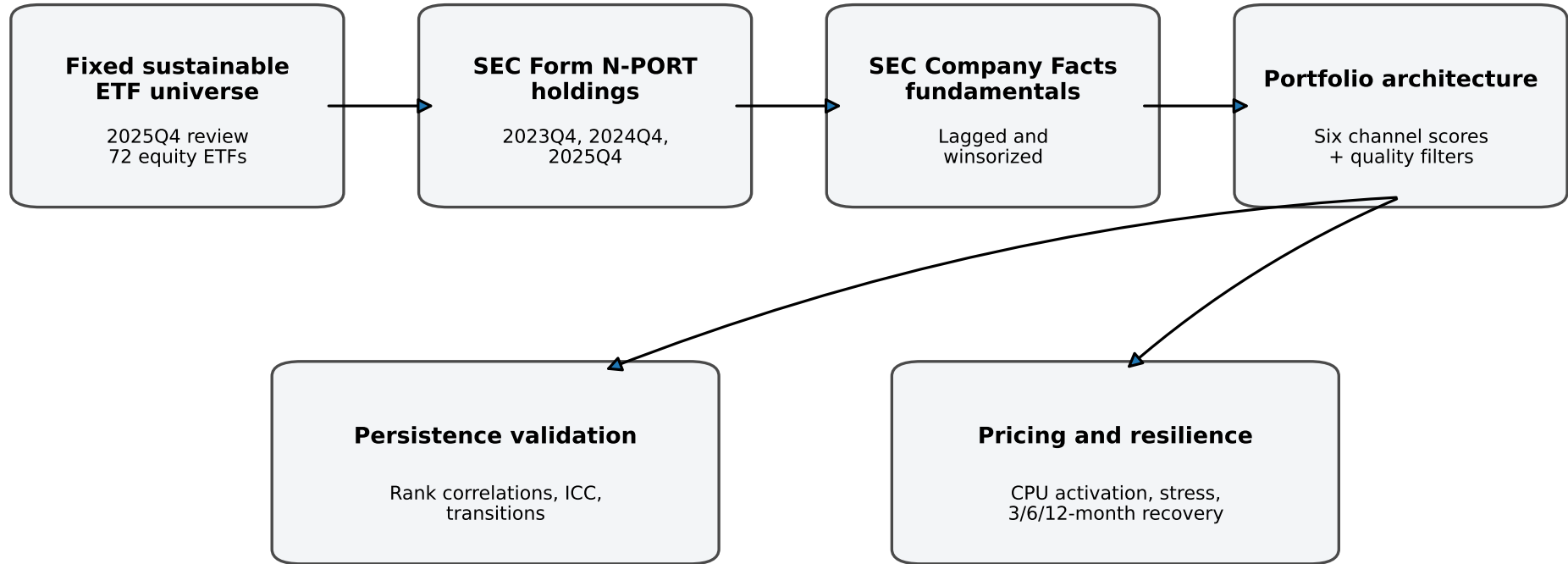


Figure 1. Empirical data architecture



Note. Mandatory N-PORT holdings are matched to a fixed fund universe and lagged SEC Company Facts. The resulting architecture is validated across reporting dates before pricing and recovery tests are estimated.